MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5678

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, subminiature, sharp cutoff

Outline	---	2-1 (EIA)
Base	---	5 inline leads, tinned
Envelope	---	T2X3, with metallic shield coating
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5
Element	---	a Red dot	g2	-f sd 1g3 (see note 1)	g1	+f (see note 1) 2g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Rg1	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	1.50	100	---	75	---	See note 6
Minimum:	1.00	---	---	---	---	---
TEST CONDITIONS:	1.25	67.5	0	67.5	5	---

GENERAL:

Qualification - Required

(E) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:09Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (anode)		0.65	II	Ib	1.2	2.4	mAdc
(E) 1266	Total grid current	Eb = 80 Vdc; Ec2 = 80 Vdc; Ec1 = -1.0 Vdc; Rg1 = 0.1 Meg (max) (see note 2)	0.65	II	Ic1	0	-0.5	µAdc
1306	Transconductance (2)	Ef = 1.0 Vdc (see notes 2 and 3)	0.65	II	Sm	600	1,450	µmhos
1201	Tap shorts		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1256	Electrode current (screen)		---	---	Ic2	0.35	0.7	mAdc
1116	Lead fatigue (subminiature)		4.0	S3	---	4.0	---	arcs
1031	Low-frequency vibration	Rp = 10,000 ohms	---	---	Ep	---	400	mVac
1331	Direct-interelectrode capacitance	Without shield	6.5	Code E	Cgp	---	0.01	pF
		Without shield			Cin	3.0	4.4	pF
		Without shield			Cout	3.7	5.5	pF
1306	Transconductance (1)	See note 3	---	---	Sm	750	1,450	µmhos
1306	Transconductance (3)	Ef = 1.0 Vdc (see note 3) take reading after 15 minutes	4.0	S3	Sm	600	1,450	µmhos
1016	Moisture resistance	See notes 4 and 5	---	---	---	---	---	---
1301	Filament current		---	---	If	44	56	mAdc
(E) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A	---	---	---	---	---	---
---	Life-test end point (500 hours)	Transconductance (1)	---	---	Sm	550	---	µmhos
	Periodic-check test							
(E) 1202	Filament burnout		---	---	---	---	---	---

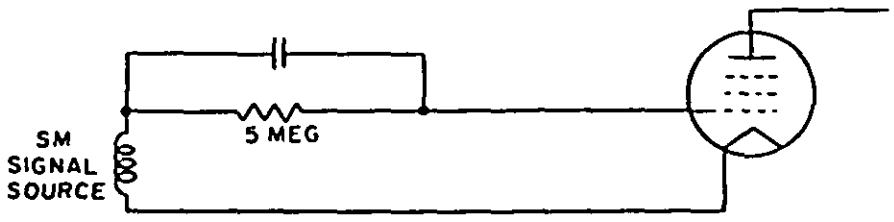
NOTES:

1. Grid (3) is composed of two separate deflector plates, one of which is connected to pin 3 and the other to pin 5.
2. This test to be performed at the conclusion of the holding period.

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:09Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

(E) 3. Test in circuit:



Bypass capacity shall have a reactance of less than 20,000 ohms at the test frequency.

4. At the completion of the moisture resistance test, and after a 2-hour holding period, there shall be no evidence of impairment of the coating after rubbing with a soft chamois held firmly in the hand. Exerting as much pressure as possible, the metallic coating shall be rubbed 12 times. Impairment of the coating in the form of blisters, cracks, and peeling shall be cause for rejection. Ten cycles of method 106 of MIL-STD-202, shall be performed.

(E) 5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, AQL 6.5 percent, shall apply.

6. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 80
DSA - ES

User activities:
Army - WC, ME
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5960-2401-36)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:09Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-TD-SS
Fort Monmouth, New Jersey 07703

To detach this form, cut along this line

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:09Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/281E, ELECTRON TUBE, RECEIVING TYPE 5678

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:09Z
Check the source to verify that this is the current version before use.